

# Tenkawa Village — "Making Use of Forests, Protecting the Land"

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## New ideas are changing the region

2025/09 revised edition, Ver.01 — Tenkawa Village (天川村)

English translation of [202509活かす森林\\_改訂横版.pdf](#), sent by Sugimoto Kazuya (杉本和也) on 2026-08-25 following the Tenkawa village visit. Source is a Tenkawa Village outreach brochure; page numbers below match the original.

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## Cover

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Forests cover 97% of Tenkawa Village's land. In a village where forests and forestry so heavily shape local conditions, the old uniform policy — planting sugi and hinoki and aiming mainly at timber production — leaves the village exposed to unstable factors like timber prices and workforce shortages. The goal now is to draw out the full multifunctional value of the forest while sustaining an attractive, environmentally sound, safe and comfortable life, and a stable village economy and employment. Tenkawa is pursuing a range of projects toward a mountain village that lives sustainably with its forest.

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## p.001 — Foreword

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Tenkawa Village once relied on forestry as its main industry, from harvesting vast natural forests through to afforestation and stand-tending under expansion-forestry policy, and for a time it also actively produced polished logs (磨き丸太) as a local industry. But as planted trees matured in recent years, a collapse in timber prices and shifting construction styles and values cut demand for wood. Forest owners and forestry workers lost economic expectations from their forests, interest and motivation declined, and the resulting drop in the number of people working in forestry has driven depopulation — a major blow to the village.

Yet 97% of the village's land is forest, and it cannot simply be left unmanaged given its role in the environment and disaster prevention. How to keep the interest of forests and forestry alive as a contributor to national land conservation, and how to turn that into a strength for the village, has become a major policy challenge. Taking the 2017 renewal of the village's hot-spring bathing facility as a turning point, the village introduced a firewood boiler and began using forest residue and previously unused timber — left abandoned until then — as fuel. Rather than the pellets or chips commonly used for woody-biomass utilization, the village focused on "firewood" (薪), which can be processed within the village and requires less processing energy. This succeeded in lowering production costs, enabling in-village processing, and building a village economic cycle in which residents casually bring wood to the plant and are paid for it with a regional gift certificate. This effort, we believe, has led forest owners to a major realization about the diversity a forest holds.

There has long been a trend of urban residents and companies supporting mountain villages and forests as part of their own image strategy. Today, though, we feel the support we receive is no longer just strategic branding but comes from people who truly understand the importance of forests and sense the crisis that exchange between mountain and city areas must be made one of Japan's most robust foundations for the future. We in the mountain villages must move past the old era in which forests were valuable mainly through high-value timber transactions, and — on the premise of national land conservation — even through small

efforts, send out from this village to the whole country our thinking on the future shape of forests, forestry, and mountain villages.

— Kurumatani Shigetaka, Mayor of Tenkawa Village

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## p.002 — Aiming for a village where people can go on living with the forest, forever

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### Life in the village

97% of Tenkawa's land is forest. The village holds the Yoshino-Kumano National Park and the World Heritage Omine Okugake-do pilgrimage route, and is rich in nature and history. Keeping this appealing village shining forever depends on the lives of the people who continue to live here. The village needs both to protect a safe, comfortable life and to make rich use of its forest resources as a stable livelihood.

### Forest resource utilization

Until now, use of forest resources centered on "timber" (building material). In recent years, prolonged low timber prices have depressed forestry management motivation. Tenkawa Village is therefore shifting away from a timber-only view and aiming to make full use of forest ecosystem services:

- Utilization of local in-region energy
- Revitalizing local industry through planting medicinal trees
- Utilizing aroma, herbs, and similar products
- As a greenhouse-gas absorption source
- Using natural and cultural landscape as a tourism resource

### Forest conservation

Alongside declining motivation to manage forestry, more forest is being left unmanaged and falling into disrepair. The village is thinking about a healthy forest cycle that makes use of grown, enriched forest resources while pursuing the fullest expression of their multiple functions:

- Creating a cycle through resource utilization
- Attracting corporate support for the forest
- Converting neglected plantation forest to mixed forest

*(Diagram: "Life in the village" balances on a scale between "Forest resource use" and "Forest conservation," resting on the base "97% forest — Tenkawa Village.")*

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## p.003 — Actively utilizing woody biomass

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As timber prices fell, forest owners' motivation to manage their forestry declined, and forest within the village increasingly went untended; even where thinning was carried out, the cut wood was often simply left behind rather than used as a resource. To break this pattern, the village focused on using this kind of forest residue as village energy, and began utilizing woody biomass. It is now taking root as a contributor to forest-environment

conservation and the village's internal economic cycle.

## "Firewood" — easy to process into fuel

The most common approach to woody-biomass use elsewhere is wood pellets or chips, whose handling and combustion control are relatively easy. Tenkawa Village instead chose "firewood" (薪), the easiest to process and store as fuel. Firewood requires little processing labor, no large initial capital cost for processing equipment, and no cost for transport or packaging — enabling low-cost woody-biomass energy use.

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## p.004 — Buying logs with a regional gift certificate / Resource use through firewood boilers at 4 village facilities

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### Buying logs with a regional gift certificate

Logs collected within the village are basically brought in cut to 1m lengths. At 1m, unless the diameter is unusually large, a log can be carried by hand and transported in a light truck bed. The village buys these logs at a rate of ¥7,000 per cubic meter, paid in a regional gift certificate that can be used at shops within the village. The firewood is then sold to the village's bathing facility and elderly-welfare facility at ¥12,000 per cubic meter. Because the gift certificates are also usable at fuel retailers in the village for gasoline or kerosene, there's no friction with local retailers. The certificates can, of course, also be used for other food and daily goods, contributing to the village's internal economic cycle.

*(Diagram: "Economic cycle in Tenkawa Village created by forest management" — Unmanaged forest → Villagers (thinned logs) → Village shops / Bathing facility / Elderly welfare facility, connected via the regional gift certificate, run through the "Tenkawa Village Forest Power Council.")*

### Resource use via firewood-boiler adoption at 4 village facilities

Tenkawa Village uses this firewood as a heat source for public facilities. Use began in Heisei 29 (2017) at "Amanogawa Onsen." In Reiwa 3 (2021) the small-scale multifunctional long-term-care facility "Momiji-no-Sato" joined, in April Reiwa 6 (2024) the "Dorogawa Onsen Visitor Center" opened, and during Reiwa 7 (2025) "Mizuha-no-Yu" will also finish conversion to a firewood boiler. Total annual firewood consumption across all facilities will reach about 1,200 cubic meters. Switching from heavy fuel oil to firewood cuts about 170,000 liters of heavy-oil use per year, achieving roughly 430 t-CO<sub>2</sub> of emissions reduction.

In addition, because this woody-biomass project keeps everything — felling, processing, and consumption — moving within the village, CO<sub>2</sub> emissions from transport are also greatly reduced.

*(Map showing the 4 facility locations: Momiji-no-Sato, Dorogawa Onsen Visitor Center, Amanogawa Onsen, Mizuha-no-Yu.)*

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## p.005 — Building a regionally rich forest centered on Kihada (Amur cork tree)

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Forest development centered on kihada (キハダ, Amur cork tree) began after felling on a prefectural joint-profit forest (分収林) in Tenkawa's Dorogawa district. A 120-year surface-rights lease had expired, the whole area was clear-felled at once, and the felled land was returned. After felling, the landowner's share of the proceeds was small, and replanting looked difficult in practice — but this became the occasion for a decision to bring back a regionally rich forest instead, an effort that has drawn support from a number of companies.

### Sugi and hinoki again — or not?

Beyond the lack of clear funding for replanting, there was also the question of whether to plant and grow sugi (cedar) or hinoki (cypress) here again. Under the expansion-forestry policy once pushed nationwide, Tenkawa Village too planted sugi and hinoki across almost all its mountain forest, except in areas such as National Park special protection zones where planting is legally restricted. The village benefited considerably from that policy through employment and revenue, but today, as forest owners' management motivation has declined and "abandoned plantation forest" has increased, Tenkawa — which experienced disaster in the great flood of Heisei 23 (2011) — has come to question a sugi/hinoki-only forest policy, from both a disaster-prevention and an environmental standpoint. The village's Dorogawa district is blessed with history, culture, and scenery and is one of the Kinki region's leading tourist destinations. Taking this felling as the opportunity, the village decided it wanted to bring back a forest where the changing of the four seasons can be felt in this area, and hand it down to the children and grandchildren who will go on living here.

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## p.006 — Forestry working together with local industry

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Tenkawa's Dorogawa district has a 1,300-plus-year history as the lodging town (宿場町) for pilgrims to the Shugendo sacred mountain Omine-san. The Japanese-and-Chinese herbal medicine "Daranisuke-gan" (陀羅尼助丸), which ascetic practitioners always carried, shares that history and is still manufactured and sold today as the area's specialty product. However, kihada bark's inner layer, "oubaku" (黄檗) — the raw material — has become harder to obtain as conditions around the region have changed, including through expansion forestry. Anticipating future depletion of both domestic and overseas supply, the village decided to begin developing a regionally rich forest centered on kihada, with the goal of creating growing conditions so the raw material itself can also be sourced locally.

Because the village was looking for a subsidy program that would place as little burden as possible on the village, the forestry project's partnership with local industry was recognized, and it was selected for support from the "Agriculture, Forestry and Fisheries Future Fund" (農林水産業みらい基金), receiving about ¥60 million in project costs over 3 years (90% subsidy rate).

*(Captions: Japanese-Chinese herbal medicine "Daranisuke-gan" of Mt. Omine; PR poster of the Agriculture, Forestry and Fisheries Future Fund; "Forest-building starts from growing locally sourced seedlings"; "Work roads opened at 350m/ha network density"; "20,000 broadleaf seedlings planted with local cooperation.")*

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## p.007 — A connection that grew out of forest-building

### The encounter with more trees

After the 3-year Mirai Fund project ended, an encounter arose with the general incorporated association "more trees," which had taken an interest in Tenkawa Village's efforts. This immediately led to a tripartite "Forest Conservation Agreement" among the village, more trees, and the local zaisanku (財産区, village property ward), making it possible for our regionally rich forest-building to move forward with this powerful partner toward a shared vision.

more trees (一般社団法人 more trees) works with regions to conserve forest, running "more trees forests" (more treesの森) around the world under the keyword "connecting cities and forests," aiming for a society where people live together with forest, always.

After the agreement was signed, more trees called on numerous urban companies to take part in forest-building in Tenkawa's Dorogawa district, and six companies now support our forest-building. We deeply appreciate that deepening exchange and mutual understanding between cities and mountain villages, between large corporations and mountain-village primary industry, is among the most important activities for keeping this country sustainable going forward, and that they have taken up that activity. We intend to keep expanding and enriching these efforts and to keep working boldly so we can go on sending them out to the whole country.

(Caption: "Mayor Kurumatani and Secretary-General Mizutani exchanging the signed agreement.")

### Estimated project cost of the kihada forest-building plan for the Dorogawa felled area (2019–2026), ¥ millions

|                                     | 2019 | 2020 | 2021 | 2022 | 2023 | 2024 | 2025 | 2026 | Total |
|-------------------------------------|------|------|------|------|------|------|------|------|-------|
| Agri/Forestry/Fisheries Future Fund | 18.0 | 18.0 | 18.0 | –    | –    | –    | –    | –    | 54.0  |
| Operator's own contribution         | 2.0  | 2.0  | 2.0  | –    | –    | –    | –    | –    | 6.0   |
| National subsidy                    | –    | –    | 1.4  | 4.1  | 7.0  | 6.7  | 6.9  | 4.4  | 30.5  |
| more trees corporate grants         | –    | –    | 1.5  | 9.0  | 12.4 | 15.6 | 13.3 | 11.0 | 62.8  |
| Total                               | 20.0 | 20.0 | 22.9 | 13.1 | 19.4 | 22.3 | 20.2 | 15.4 | 153.3 |

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## p.008 — Sponsoring companies

(Photo spread: volunteers at "UGGの森" (UGG's Forest, supported by Deckers Japan G.K. via more trees) and "TRACEの森" (TRACE's Forest) planting events; project-plan/track-record supporting-company logos: DECKERS BRANDS, SMBC三井住友銀行 (Sumitomo Mitsui Banking Corporation), SMBC三井住友カード (Sumitomo Mitsui Card), SMFL, TRACE, waternet; further photos of a group at "みらい2030の森" (Mirai 2030 Forest) and a snow-covered "三井住友カードの森" (Sumitomo Mitsui Card Forest) sign.)\*

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## p.009 — J-Credit initiative

### Nara Prefecture's first! Registered forest-management project

The "J-Credit Scheme" has been slow to take off even in forest-rich Nara Prefecture, but Tenkawa Village decided to "just try it anyway," and in FY2022 (Reiwa 4) took on registering a forest-management project on one piece of village-owned forest, named "Towa (Eternal) Earth Project" — achieving Nara Prefecture's first such registration.

### The key is proper forest management

Certification, registration, and crediting under the J-Credit scheme depend above all on "proper management based on a forest plan." In Tenkawa Village, credits are earned through forest management carried out by the local regional-revitalization corps (地域おこし協力隊) active in village-owned forest. The target forest area is a modest 32ha, and absorption is also around 150 t-CO2/year (roughly 2,000 t-CO2 total over the project period), but as of FY2025 (Reiwa 5), sellable credits have also been secured.

### Using the created credits to communicate the mountain village's importance

Rather than the size of the sales revenue, the village sees the credits it creates as more valuable for how they let many companies broadly recognize their contribution to Tenkawa Village's forest and global environment. Credits through FY2025 (Reiwa 7) are already sold out, and are being used by buyers for carbon offsetting in paper and similar manufacturing. As products offset with Tenkawa Village's credits are used widely in society, the village hopes to raise broader awareness of its forests' greenhouse-gas absorption capacity and to continue communicating the importance of mountain villages, forests, forestry, and global environmental conservation.

### Tenkawa Village "Towa (Eternal) Earth Project" — net absorption (t-CO2)

| FY   | Net absorption (t-CO2) |
|------|------------------------|
| 2022 | 139                    |
| 2023 | 152                    |
| 2024 | 152                    |
| 2025 | 142                    |
| 2026 | 128                    |
| 2027 | 122                    |
| 2028 | 122                    |
| 2029 | 122                    |
| 2030 | 119                    |

| FY           | Net absorption (t-CO2) |
|--------------|------------------------|
| 2031         | 114                    |
| 2032         | 112                    |
| 2033         | 112                    |
| 2034         | 112                    |
| 2035         | 107                    |
| 2036         | 104                    |
| 2037         | 102                    |
| 2038         | 8                      |
| <b>Total</b> | <b>1,969</b>           |

(Caption: "Regional-revitalization corps members sweating over forest-management work.")

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## p.010 — Next-generation forest and its stewards

The project moved from the Mirai Fund (2019) into more trees' forest-building (2021), with a plan to keep planting continuously on the large felled area through 2026. But to make the fullest, most circular use of forest resources, the village is also thinking about the next step for forest-building.

### Mixed conifer-broadleaf conversion for the fullest expression of forest function

About 70% of Tenkawa's forest is sugi/hinoki plantation. Some of this plantation, planted on suitable ground and well-tended, can be expected to yield good-quality timber, but much of it was planted on unsuitable ground, has had delayed stand-tending, and cannot be expected to produce good-quality timber. Momentum is building not to leave such plantation forest untended indefinitely, but to gradually bring it back closer to a natural regional forest. Because Tenkawa Village has an established system for using local timber as woody biomass, even lower-quality wood reliably becomes local work once it is felled and hauled out. This effort is key to raising forestry again as the mountain village's core industry. And after felling, planting a variety of species will restore a forest rich in regional character.

### Developing and organizing the next generation of stewards

However fine a plan may be, forest-building does not move forward without people to carry it out locally. Since Reiwa 1 (2019), Tenkawa Village has recruited and hired regional-revitalization corps members for forestry every year. By the end of FY2024 (Reiwa 6), 11 people had been hired, 7 had completed their terms, and 4 of those remain engaged in forestry in Tenkawa. In addition, local peers who also became interested in forestry newly established a limited liability partnership (LLP), putting in place a clear organizational structure for carrying out the work. The village intends to keep expanding this kind of development and organization of stewards village-wide.

*(Captions: "Tohoku University Professor Emeritus Kiyowa Kenji [seiwa Kenji] discussing the plan on-site"; "The LLP that a group of local young people, awakened, organized in spring 2024.")*

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## **p.011 — Supporting motivated operators and cultivating love for the region**

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### **Supporting work roads and protective gear with the forest environment tax**

Advancing forest-resource use requires work roads, but in Tenkawa's steep terrain, a makeshift work road can itself trigger disaster. A work road built to last is the ideal, but building one costs money. The village is therefore using the forest environment tax (森林環境税) to support the development of durable work roads. A road built to last also plays a major role in later planting and stand-tending after logs are hauled out. Sourcing chainsaws, protective gear, and other safety-standard equipment is also a major hurdle for young entrants to the field — this too is supported through funds from the forest environment tax, contributing to safer forestry work going forward.

### **Making a beautiful village even more beautiful, through the region's own effort**

About 30% of Tenkawa Village lies within the national park, and with the World Heritage Omine Okugake-do pilgrimage route as well, it is one of the Kansai region's leading tourist destinations — the village takes pride in being beautiful. To hand this beautiful village down to the next generation even more beautiful, the village has begun activities that bring the community's efforts together.

The Dorogawa Onsen Visitor Center, which opened in April Reiwa 6 (2024), finished surrounding parking-lot improvement works in March Reiwa 7 (2025), completing the setup to properly welcome guests. This led to a plan to create a ring of Japanese maples in the "Menmera (Our) Forest" beside the Center, and administration and local residents together sweated planting maple seedlings. Going forward the village wants to continue such activities that rise up from the community rather than only being administration-led, and hand down a beautiful village, made even more beautiful, to the next generation.

*(Captions: "A durable work road resistant to washout is essential for using and protecting forest resources"; "A beautiful village is residents' pride, made more beautiful through the region's own strength.")*

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## **p.012 — Making the most of a mountain village's strengths, for a life only possible here**

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### **Forest work that appeals to women too**

The survival of a mountain village hinges on young women. Creating enjoyable work that women can take pride in while living in the village is very important for a mountain village. Since 2019, Tenkawa Village has been feeling its way toward making rich use of its forest resources as aroma and herb products. There is also a historical culture of using forest resources such as medicinal trees and herbs, as in the "Daranisuke-gan"

tradition, and in Tenkawa, using forest resources for mind and body wellness comes very naturally as a local industry. The village has developed and sells products using essential oils from kuromoji (spicebush), nioikobushi (fragrant magnolia), koyamaki (Japanese umbrella pine), sugi, and hinoki. Going forward, the village wants to keep growing this into an appealing local industry, with women discussing product development and sales approaches together.

## **A mountain village where young people can enjoy life**

Mountain villages face major challenges like population decline and aging, but with a 97% forest ratio, Tenkawa Village is an ideal environment for exploring new ways of living that make use of the forest. In terms of woody-biomass use it truly is a "treasure mountain," and corporate forests, regionally rich forest-building, and J-Credit creation all give young people involved a sense of dream and fulfillment. Being at the leading edge of new technology adoption is also expected to shift the initiative to young people and revitalize forestry work. Conserving a community-led landscape nurtures a sense of belonging and pride locally. Creating an employment environment where young women can work is essential to the village's survival. Going forward, alongside creating this kind of forest-based work, the village will also keep advancing improvements to the living environment. Even within the constrained environment of a mountain village, Tenkawa Village will make use of its forest to protect the village and protect the nation's land.

*(Captions: "Forest resource use connects to new dreams"; "A fulfilling life only possible here.")*

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